

1 Introduction

Predicting the properties of crystals is a problem with many useful applications, such as understanding the behavior and functionality of crystalline solids (Tao et al., 2021; Parikh et al., 2022). Additionally, the ability to predict crystal properties would greatly accelerate the discovery and development of new crystals by identifying candidate materials warranting experimental study (Meredig et al., 2014; Oliynyk et al., 2016; Raccuglia et al., 2016; Ward et al., 2016; Chen et al., 2019; Choudhary and DeCost, 2021; Chen and Ong, 2022).

Similarly to proteins (Ioannidis et al., 2019; Strokach et al., 2020; Jiang et al., 2020; Shen et al., 2021; Wang et al., 2022; Jha et al., 2022; Réau et al., 2023), crystals are often represented as graphs that model the interactions between nearest neighbors; atoms in atomic crystals and molecules in molecular crystals (Schütt et al., 2017; Xie and Grossman, 2018; Chen et al., 2019; Choudhary and DeCost, 2021; Chen and Ong, 2022). For either case, crystal lattice sites are represented as nodes and the bonds (e.g., ionic or covalent or van der Waals) between them are represented as edges. Graph neural networks (GNNs) are then typically used to learn the contextual representation of each node and edge within the crystal graph to predict its properties. However, GNNs face several challenges to effectively predict crystal properties: 1) The need to efficiently encode the periodicity inherent to any crystal which results from the repetitive arrangement of unit cells within a lattice, a representation distinct from standard molecular graphs (Yan et al., 2022). 2) The complexity of incorporating critical atomic and molecular information such as bond angles (Choudhary and DeCost, 2021) and crystal symmetry information such as space groups (Kaba and Ravanbakhsh, 2022). Finally, 3) graphs may lack expressiveness, which is useful in conveying complex and nuanced crystal information that is critical for accurate crystal property prediction.

The aforementioned challenges are due to the complexities of crystal graph representation. We aim to mitigate these challenges by modeling the crystal structure from its text description as opposed to its graph representation. Textual data contain rich information and are very expressive; additionally, incorporating critical/desired information in text is generally more straightforward compared to graphs. Crystal representations can for instance be learned by pretraining large language models (LLMs) on a large body of scientific literature which contains diverse chemical and structural information about crystal design principles and fundamental properties (Beltagy et al., 2019; Walker et al., 2021; Huang and Cole, 2022; Gupta et al., 2022). Then, one can finetune these pretrained LLMs on labeled data to solve specific tasks such as crystal property prediction (Korolev and Protsenko, 2023; Qu et al., 2023), crystal recommendation and ranking (Qu et al., 2023), and synthesis action retrieval (Song et al., 2023). However, these methods also face the challenges of limited pretraining and downstream data, limited computational resources, and a lack of efficient strategies to use the available resources.

In this work, we demonstrate effective strategies for the use of LLMs for the accurate prediction of crystal properties from their text descriptions. Existing approaches for finetuning LLMs for prediction tasks often either rely on both the encoder and the decoder or leverage encoder-only large models that tend to have as many